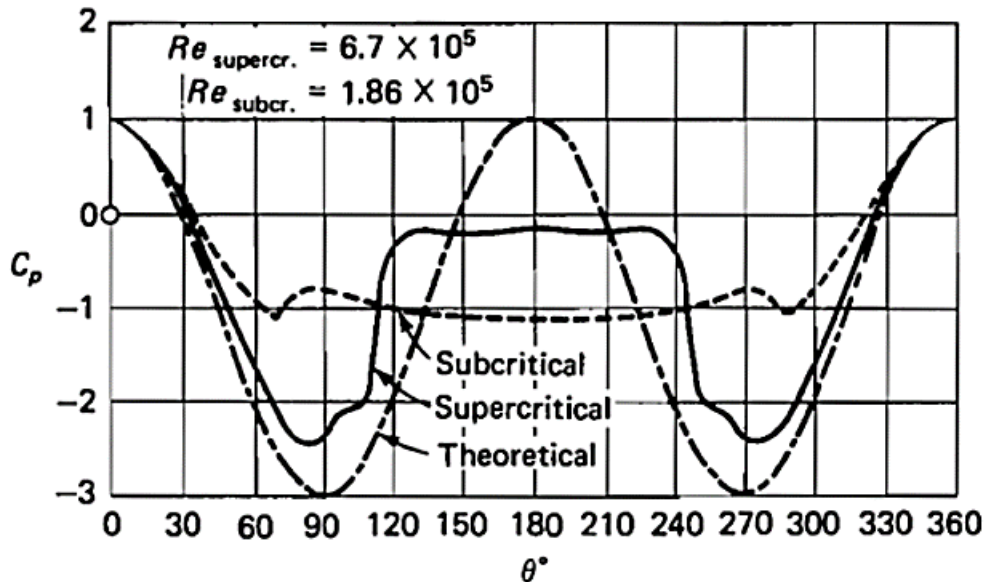


- 1- The pressure coefficient distribution on a cylinder is shown from three different distributions in the figure below.



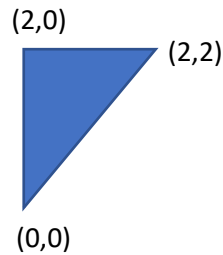
- Determine the potential, laminar, and turbulent flows on the graphs shown in the figure.
- Explain the reason for constant pressure regions in the two distributions shown. Why is the location of the constant pressure region different in these two distributions?
- At a 30-degree angle, calculate the pressure coefficient of the potential flow using its relation and explain what does this value implies.
- Before the separation point, the potential, laminar, and turbulent solutions do not necessarily coincide. Explain the reason for this phenomenon.
- Based on the figure, compare the pressure in the laminar and turbulent separation regions (Hint: Consider the diameter of the cylinder, viscosity, pressure, and air density to be the same in both regimes, and use the given Reynolds number to calculate the velocity).

2- A two-dimensional incompressible flow is described by the stream function $\psi = x^2 - y^2$. Find the value of $\frac{\nabla p}{\rho}$ at the point (1,2).

3- Consider the following velocity distribution:

$$\vec{v} = (x^2 + y^2) \hat{x} + 2xy^2 \hat{y}$$

- a) Is the flow rotational or irrotational?
- b) Using the line integral, calculate the circulation around the following triangle in a counterclockwise direction.



4- Calculate the circulation for the three-dimensional velocity field $V = (16x^2 + y)i + 10j + yz^2k$ over a rectangle with corners at (0,0), (10,0), (10,5), and (0,5).

5- Consider the semi-cylinder shown below. The pressure on the left side is obtained from the potential theory solution for a non-spinning cylinder, and the pressure on the right side is assumed to be the pressure at the corner, i.e., p at 90 degrees. Computer the drag and lift coefficient of this geometry.

